

CS1CA Computer Systems Architecture

Information Representation – 1

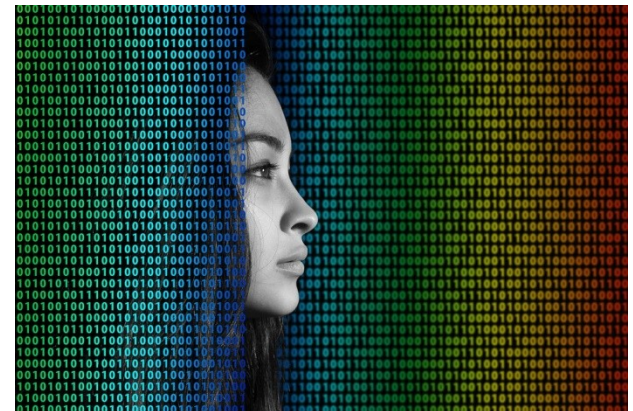
Week 1 lecture 2 of the autumn semester 2025

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Outline of the topic

- Introduction: what is information?
- Decimal, binary and hexadecimal number systems
 - Conversion between number systems
- Internal representation of characters
- Internal representation of integers
- Arithmetic and logic operations (binary)
- Internal representation of real numbers
 - IEEE 754 floating point format

This topic will be delivered in 5
lectures of weeks 1 and 2.

In this lecture

- The following contents will be delivered.
 - Introduction: what is information?
 - Decimal, binary and hexadecimal number systems
- Learning outcomes
 - To gain the knowledge of information representation in a computer.
 - To gain knowledge of the three number systems

What is information, and how to represent it in a computer?

- Information comes in various forms such as
 - Text including digits, symbols and characters
 - Visual info: images, videos...
 - Audio, temperature, ...
- Computer can only store and manipulate binary strings (in **bit patterns**).
- A conversion is needed from external to internal forms.

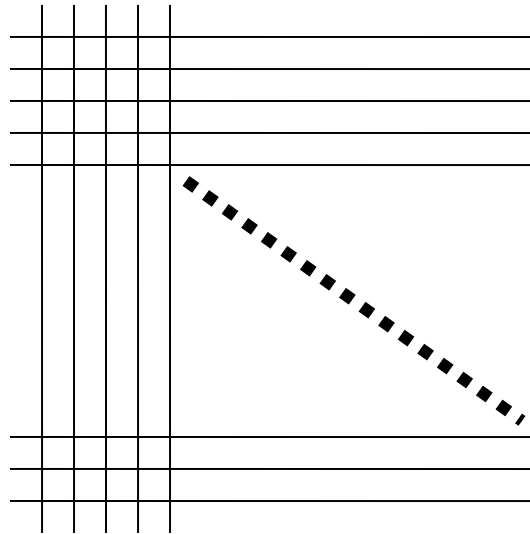
Two examples of information representation

- Text: “Hello” in ASCII

01001000 01100101 01101100 01101100 01101111

H e l l o

- Image: “a panda” in grey-level bitmap format



- Image (2D array) consists of pixels;
- Each pixel is represented by a number from 0-255 (8 bits in bit pattern).
- 0: black; 255: white.

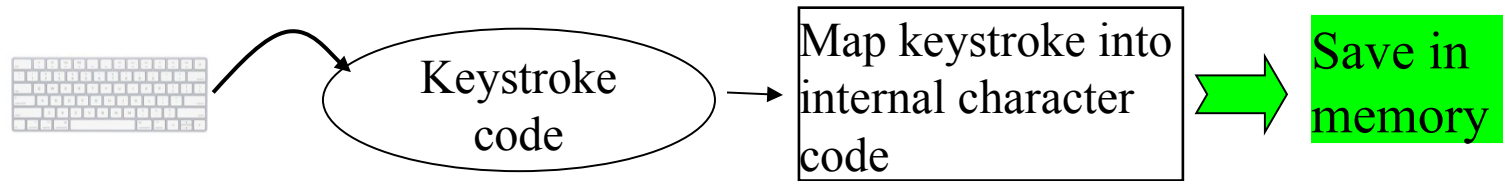
Write your name with the given code

Character	Binary Code	Character	Binary Code	Character	Binary Code
A	01000001	a	01100001	!	00100001
B	01000010	b	01100010	"	00100010
C	01000011	c	01100011	#	00100011
D	01000100	d	01100100	\$	00100100
E	01000101	e	01100101	%	00100101
F	01000110	f	01100110	&	00100110
G	01000111	g	01100111	'	00100111
H	01001000	h	01101000	(	00101000
I	01001001	i	01101001	)	00101001
J	01001010	j	01101010	*	00101010
K	01001011	k	01101011	+	00101011
L	01001100	l	01101100	,	00101100
M	01001101	m	01101101	-	00101101
N	01001110	n	01101110	.	00101110
O	01001111	o	01101111	/	00101111
P	01010000	p	01110000	0	00110000
Q	01010001	q	01110001	1	00110001
R	01010010	r	01110010	2	00110010
S	01010011	s	01110011	3	00110011
T	01010100	t	01110100	4	00110100
U	01010101	u	01110101	5	00110101
V	01010110	v	01110110	6	00110110
W	01010111	w	01110111	7	00110111
X	01011000	x	01111000	8	00111000
Y	01011001	y	01111001	9	00111001
Z	01011010	z	01111010	?	00111111
				@	01000000

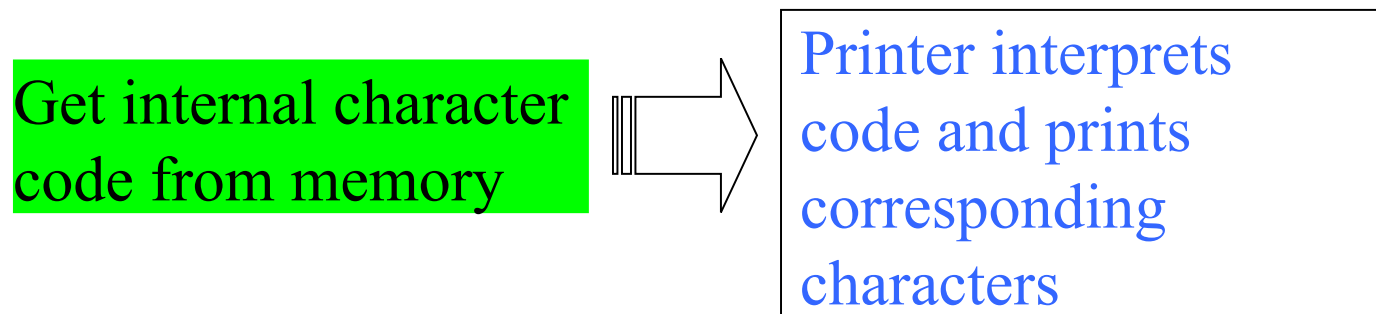
Courtesy of
the website

Getting information in and out a computer: text

- Keyboard input



- Printer output



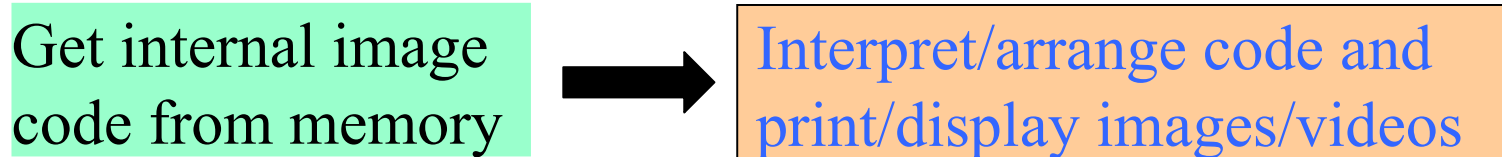
- Monitor display
 - Similar to printer

Getting information in and out a computer: images, videos, and other signals

- Camera input



- Printer/monitor output



- Other signals, such as audio, temperature, air pressure, ...

- ✓ Input: Analogue to Digital Converter (ADC) – convert analogue signals to binary representation in bit patterns.
- ✓ Output: Digital to Analogue Converter (DAC) – convert digital signals to analogue signals (may not be necessary).

Leading to number systems

- Numbers are input from the keyboard and output to the printer/monitor as a string of characters.
- Programming languages, such as C/C++, Java, Python, Basic, etc. provide a library of procedures that handle the conversion of external character code strings to internal number bit strings and vice versa.
- Both integers and decimal real numbers need to be represented internally as **fixed-length bit patterns** in a correct format. How to represent them?

Mathematical background of number systems and conversions (1)

- **Integer division** $A \div B$ delivers two results: **quotient** q and **remainder** r

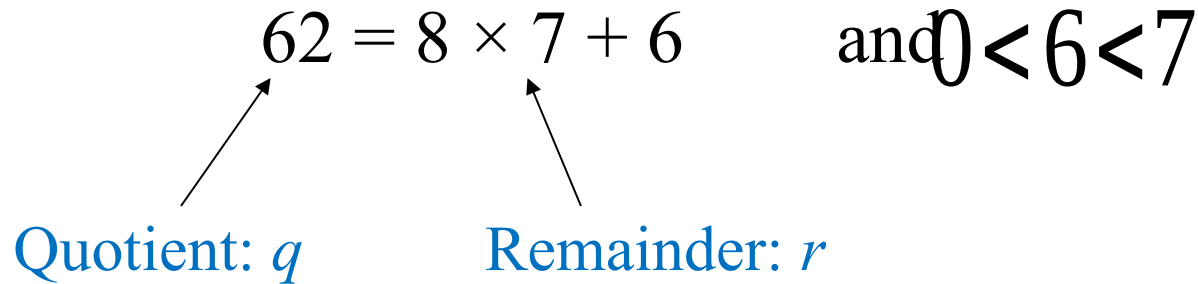
$$A = q \times B + r \quad \text{and} \quad 0 \leq r < B$$


quotient remainder

- **div** and **mod**: two operations in use
 $A \mathbf{div} B = q$ and $A \mathbf{mod} B = r$

Example – 1

- $A=62, B=7$. $62 \div 7$ delivers two results: quotient **8** & remainder **6**

$$62 = 8 \times 7 + 6 \quad \text{and} \quad 0 < 6 < 7$$


Quotient: q Remainder: r

- **div** and **mod**: how to find q and r
 $62 \text{ div } 7 = 8$ and $62 \text{ mod } 7 = 6$

Mathematical background of number systems and conversions - 2

- **Elementary polynomials**: a function of powers of x , e.g. a quadratic $6x^2 + 3x^1 + 9x^0$
- There are both positive and negative powers of x ;
- Polynomial representation in general is given as below.

- $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x^1 + a_0 x^0 + a_{-1} x^{-1} + \dots + a_{-m} x^{-m}$
The coefficients a_i ($i = -m, \dots, -1, 0, 1, \dots, n$) are constant.

Example – 2

- A real number 3769.54
- It can be represented by a polynomial as
$$3 \times 10^3 + 7 \times 10^2 + 6 \times 10^1 + 9 \times 10^0 + 5 \times 10^{-1} + 4 \times 10^{-2}$$
- We can see that the power to 10 can be positive, zero, or negative.
- The coefficients are constants corresponding to the number. Later you will see that they are **the remainders** in number conversion processes.

Numbers to any base R

- A real number can be expressed as

$$N_R = r_n R^n + r_{n-1} R^{n-1} + \dots + r_1 R^1 + r_0 R^0 + r_{-1} R^{-1} + \dots + r_{-m} R^{-m}$$

- The coefficients are in the range of $0 \leq r_i \leq R-1$

- We use the following way to write a number

$$N_R = \langle r_n r_{n-1} \dots r_1 r_0 \cdot r_{-1} r_{-2} \dots r_{-m} \rangle_R$$

- For example: $\langle 1011 \rangle_2$, $\langle 368 \rangle_8$; (any issue?)
- The braces $\langle \rangle$ are only used to distinguish the base R from other subscripts.

$$\langle 367 \rangle_8 = 367_8, \text{ and } \langle 1011 \rangle_2 = 1011_2$$

Decimal, binary, and hexadecimal number systems

- Decimal numbers have $R=10$; (r range?)
- Binary numbers have $R=2$; (r range?)
- Hexadecimal numbers have $R=16$. (r range?)
6 characters in Hexadecimal: A=10, B=11, C=12, D=13, E=14, F=15

Examples: write the following numbers in polynomial representations.

12.56_{10} , 11010_2 , $2BC_{16}$, and 1253_8

End of the lecture

Questions and exercises

- Produce a look-up table of the decimal number from 0 to 15 inclusive with their hexadecimal and binary equivalents.
- Present the following numbers in polynomial representations
(a) 3958.53_{10} (b) 17.66_8 (c) 1100.011_2 (d) $BA19_{16}$
- A colour image has a size of 100×150 (100 pixels in each column and 150 pixels in each row, *i.e.* 100 rows and 150 columns in size). Each pixel has three channels, *i.e.* red, green, and blue, which take 3 bytes (1 byte = 8 bits).

Calculate how many bits are needed to represent the image.